Reg. No. | | | | | | | | | | |

[14AE34]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER–2015

FLUID MECHANICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer any **FIVE** full Questions, choosing atleast two questions from each part.
2. Assume any data necessary and indicate.

PART - A

1. a. Distinguish between the following and mention their units. **10**
   - i. Specific weight and mass density.
   - ii. Surface tension and capillarity.
   - iii. Dynamic viscosity and kinematic viscosity.

   b. Obtain an expression for capillary rise. **04**

2. a. State Pascal's law and show that $p_x = p_y = p_z$ **08**

   b. An open tank contains water up to a depth of 2m and above it an oil of sp.gr.0.9 for a depth of 1m. Find the pressure intensity **06**
   - i. at the interface of the two liquids
   - ii. at the bottom of the tank.

3. a. Obtain the expression for the meta-centric height of a floating body using analytical method? Explain with a neat sketch. **08**

   b. A solid cylindrical of diameter 4m has a height of 4m. Find the meta centric height of the cylinder if the specific gravity of the material of cylinder is 0.6 and it is floating in water with its axis vertical. State whether the equilibrium is stable or unstable. **06**

4. a. Derive an expression for continuity equation in three dimension. **06**

   b. Derive an expression for continuity equation in polar co-ordinate for two-dimensional steady incompressible flow. **08**

PART - B

5. a. Derive an expression for Euler's equation of motion. **07**

   b. The water is flowing through a taper pipe of length 100m having diameters 600m of the upper end and at the lower end, at the rate of 5 litres/s. The pipe has a slope of 1 in 30. Find the pressure at the lower end if the pressure at the higher level is 19.62N/cm^2. **07**

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6. a. With an example explain the principle of dimensional homogeneity. Explain Buckingham PI theorem. **04**

b. Given that force $F = fn(L, V, \rho, \mu)$, Where F is force, L is length, V is velocity, μ is coefficient of viscosity and ρ is density of fluid. Derive the non-dimensional parameters using Buckingham PI theorem. **10**

7. a. Derive an expression for shear distribution and velocity distribution in a pipe. **08**

b. A crude oil of viscosity 0.97poise and relative density 0.9 is flowing through a horizontal circular pipe of diameter 100mm and length 10m. Calculate the difference of pressure at the two ends of the pipe, if 100 kg of oil is collected in a tank in 30 seconds. Assume Laminar flow. **06**

8. a. Obtain expressions for total drag and lift force. **06**

b. Explain boundary layer concept. **08**